

Department of English

End Semester Examination

Spring Semester, 2021

(For the 2nd Semester Students of all the Departments of the Engineering Faculty)

ENGUGHU01: Communicative English

Full Marks: **40**

Time: **2 Hours**

Section I

1. Write short notes on **any two** of the following: (Word limit 200) 5x2
- a) Affricate and fricative sounds.
 - b) Upward and downward communications
 - c) Critical reading.
 - d) Inductive writing
 - e) Discuss the difference between intra-personal communication and inter-personal communication.

Section II

2. Answer the following questions (**any ten**): **1x10**
- a. The synonym of '**cajole**' is
 - i. (i) scold (ii) intimidate (iii) threaten (iv) persuade
 - b. Neither Jacob nor Sam deserve punishment. (Correct the sentence.)
 - c. Give one word substitute for 'that which cannot be explained'.
 - d. Make a sentence with the idiom 'dark horse'.

- e. Make a sentence with the phrasal verb 'pass out'.
- f. I bought a cycle yesterday. _____ cycle is defective. (use an appropriate article.)
- g. Write the antonym for '**inevitable**'.
 - i. unavoidable (ii) uncertain (iii) envious (iv) eatable
- h. Saket and _____ went to the movie. (Fill in the blank with relevant word: I/me)
- i. Did you _____ the rice? (Fill in the blank with relevant word: ate/eat)
- j. Though they are twins, Arnab and Gourav have _____ personalities. (Fill in the blank with relevant word: discreet / discrete)
- k. He was awarded a gold _____ for securing first rank. (Fill in the blank with relevant word: medal / meddle)
- l. He swam _____ the lake. (Use an appropriate preposition)

Section III

3. Answer **any two** of the following questions: (Word Limit – 400) **10x2**
- a) What are the barriers to communication? Discuss the means to overcome those barriers.
 - b) Comment on the various types of Non Verbal Communication.
 - c) Classify and discuss the twelve vowel sounds based on articulation with diagram.
 - d) As a marketing manager of a company write a letter to the Sales Officer of a furniture manufacturing company ordering furniture for your newly set up office.
 - e) Suppose you are an engineering student and want to participate in an inter-college paper presentation competition on 'Robotics' to be organized at NIT Durgapur. Write an email to the organizing secretary enquiring details of the competition.

Aliah University

End-Semester Examination (Spring Semester) - 2021

For 1st Year 2nd Semester B.Tech. (CEN/MEN) Programme

Paper Name: Problem For Problem Solving

Full Marks: 80

Paper Code: CSEUGES01

Time: 3 hrs

Group - A

(Answer any four questions)

4 × 5 = 20

1. Discuss about increment and decrement operators. 5
2. State the differences between procedural and object-oriented languages. 5
3. How many types of storage classes C language supports? Why do we need different types of such classes? 5
4. How is a two-dimensional array stored in memory? 5
5. What is an array of pointers? Explain with suitable examples. 5

Group-B

(Answer any six questions)

6 × 10 = 60

(All parts of the same question should be answered together)

1. (a) Discuss about contestants. 5
(b) Draw a flowchart to find the largest of three numbers. 5
2. (a) Discuss about iterative statements. 5
(b) Write a C program to print the following pattern.
1
2 1 2
3 2 1 2 3 5
3. (a) What are the advantages of using switch-case statements? 5
(b) Write a C program using switch-case statements to determine whether a given character is a vowel or not. 5
4. (a) What are functions? Why are they needed? 2 + 3
(b) Write a C program to compute the area of a rectangle using a function. 5
5. (a) How to copy the elements of one array into another array? 5
(b) Write a C program that reads an array of 100 integers and display all pairs of elements whose sum is 50. 5
6. (a) How are strings read from keyboards? 5
(b) Write a C program to count the number of each character in a string. 5
7. (a) What are arrays of structures? Give an example. 6
(b) What are the differences between structures and unions? 4
8. (a) Explain fscanf() to read data from a file. 4
(b) Write a C program to merge two files into a third file. 6

1. a) Distinguish between mesh and loop of a network. 10x2=20 Marks
- b) An Electric iron is rated 1000W, 240V. Find the current drawn & resistance of the heating element.
- c) Define i) charge ii) electric current iii) power & iv) network.
- d) Write some applications of maximum power transfer theorem.
- e) Why a single phase induction motor does not self start?
- f) Give two basic speed control scheme of DC shunt motor?
- g) Define power factor. State the conditions under which it is (i) maximum and (ii) minimum.
- h) Can DC be applied to transformers?
- i) Give some application of D.C motor?
- j) Determine the power factor of a RLC series circuit with $R=5\Omega$, $X_L=8\Omega$ and $X_C=12\Omega$.

Answer any five questions

5x12=60 Marks

2. a) Determine current in 5 ohm resistor by any one method (Ref. 1) 2x6=12 Marks

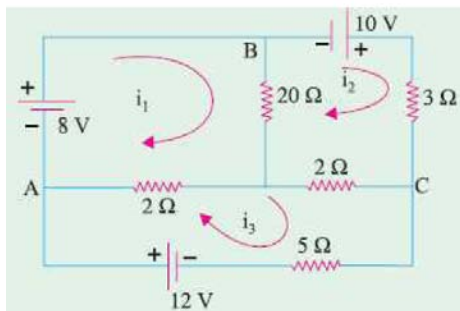


Fig. 1

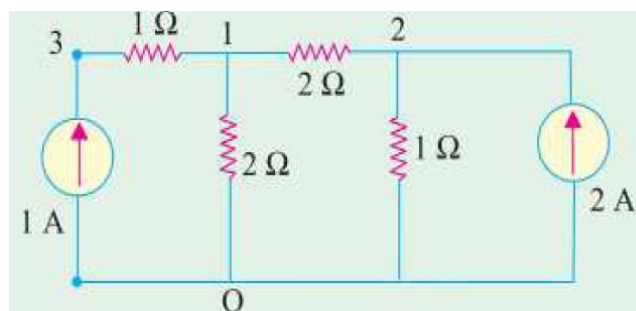


Fig.2

- b) Determine the voltages 1 and 2 of the network in **Fig. 2** by nodal analysis.
3. a) Derive the emf equation of a single phase transformer. 5 Marks
- b) A 200/50V, 50 Hz single phase transformer is connected to a 200V, 50 Hz supply with secondary winding open. (i) What is the value of maximum flux through the core, if the primary winding has 400 turns? (ii) What is the peak value of flux if the primary voltage is 200V, 25 Hz and what happens to no-load current? 7 Marks
4. a) Sketch the O.C.C of DC shunt generator. 5 Marks
- b) A series RLC circuit has 30Ω resistance, 2m H inductance and $25.1\mu\text{F}$ capacitance and is connected to a 220V, 60Hz supply. Find (i) the current, (ii) the power factor and (iii) the voltage drop across each element. 7 Marks
5. a) Draw the circuit for various types of D.C motor . 2x6=12 Marks
- b) A DC long shunt compound generator delivers a load current of 200A at 450V. The resistance of the armature, series field and shunt field are 0.03Ω , 0.015Ω and 150Ω . Calculate the emf induced in the armature. Allow a brush drop of 1V/brush.
6. a) Derive the condition for obtaining maximum power from a source to resistive load. 2x6=12 Marks
- b) What is an ideal transformer? Draw phasor diagram at different loads for an ideal transformer.
7. a) Write short notes on RMS Value of AC waveforms. 2x6=12 Marks
- b) An alternating voltage is given by $V=230\sin 314t$. Calculate i) frequency, ii) maximum value iii) average value, iv) RMS value.

ALIAH UNIVERSITY

Subject: Engineering Chemistry

Subject Code: CHMUGBS01/CH152

Time: 3 Hrs.

End Semester Examination (Semester-II; 2020-21)

Full Marks: 80

Group A (Answer any eight)

1. Write short note on: (5)
 - a. Bakelite
 - b. Teflon
2. Explain with examples: Conducting Polyaniline & GuttaPercha (5)
3. Why are silicones called Inorganic Polymers? Discuss the synthetic procedure of linear chain silicones (5)
4. (i) What is meant by green chemistry
(ii) Write informative note on the mechanism of radical polymerisation (5)
5. Write down the importance of Green Chemistry. Give the detail of the preparation of Aspirin from Salicylic acid using Green Synthesis (5)
6. Explain with examples: (i) Phenol formaldehyde resins
(ii) Microwave radiation of Green Synthesis (5)
7. How would you prepare Adipic Acid by Green Method? Write an essay on the various types of synthetic rubbers (2+3)
8. What is meant by the Vulcanization of Rubber? Write its uses (3+2)
9. What is Bio-polymers? How is it prepared? Give some importance of Bio-polymers (5)
10. Write notes on : (i) Polyvinyl chloride (PVC)
(ii) Polystyrene (PS) (5)

Group-B

1. Choose the correct alternatives for five of the following: 5×1=5
 - (i) Which of the following can be used for disinfection of water?
 - (a) Chlorine
 - (b) Hydrogen peroxide
 - (c) Ozone
 - (d) None of these

(ii) The degradation of BOD is classified as what type of reaction?

- (a) First Order Reaction
- (b) Zero Order Reaction
- (c) Second Order Reaction
- (d) Third Order Reaction

(iii) Disease associated with Mercury is

- (a) Black foot disease
- (b) Minamata disease
- (c) Itai-Itai disease
- (d) Parkinson's disease

(iv) According to recommended total hardness permissible limit, which value is acceptable for drinking water

- (a) $> 550\text{mg/l}$
- (b) $> 250\text{mg/l}$
- (c) $> 750\text{mg/l}$
- (d) $> 600\text{mg/l}$

(v) BOD is

- (a) Amount of O_2 required respiring
- (b) Amount of O_2 to decompose
- (c) Amount of O_2 to inhale
- (d) Amount of O_2 to measure COD

Group-C

4×5=20

(Short answer type questions)

Answer the following questions.

2. What is meant by hardness of water? Can hard water be used in boilers or in laundries? Justify your answer. [2+3]

3. What are the biochemical effects of Arsenic and Lead pollution? [2+3]

4. What is Dissolved Oxygen (DO)? What is its importance as water quality parameter?
[2+3]

5. In a BOD test 5

(i) Why it is necessary to cover the bottle with a stopper?

(ii) What is the necessity of diluting the waste water sample?

(iii) Why is the test run in dark (or in the dark bottle)?

(iv) What is the necessity of seeding the sample?

(v) Why the ultimate BOD (BOD_u) is not measured?

[1+1+1+1+1]

Group-D (Answer any three questions) 3×5=15

6. Prove that, the relation $BOD_t = C_0 (1 - e^{-kt})$ where the terms indicates their usual meaning.

7. Draw a flow sheet of a water treatment plant for treating surface water to confirm public water supply standard.

8. Write a short note on oil pollution in water?

9. Explain Eutrophication process?

ALIAH UNIVERSITY
DEPARTMENT OF MATHEMATICS AND STATISTICS
COURSE: ENGINEERING MATHEMATICS-II
COURSE CODE: MATUGBS02 and MA 134
Total marks : $10 \times 8 = 80$
Time allotted : 3 hours

Unless otherwise stated, notations carry their usual meanings. Candidates should read the following instructions carefully before answering the questions:

Instructions: Answer any **four** questions from part **A** and any **four** questions from part **B**.

Part-A

1. (a) If $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$, show that $A^2 - 4A - 5I_3 = O$. Hence obtain a matrix B such that $AB = I_3$.
(b) Prove that the matrix $A = \frac{1}{3} \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{pmatrix}$ is orthogonal. Utilise this to solve the following system of equations
$$\begin{aligned} x - 2y + 2z &= 2 \\ 2x - y - 2z &= 1 \\ 2x + 2y + z &= 7. \end{aligned}$$
2. (a) Solve the system of equations by matrix inversion method
$$\begin{aligned} x + z &= 0 \\ 3x + 4y + 5z &= 2 \\ 2x + 3y + 4z &= 1. \end{aligned}$$

(b) Find the rank of the rectangular matrix $\begin{pmatrix} 0 & 0 & 5 & -3 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & -7 \end{pmatrix}$.
3. (a) Use Caley-Hamilton theorem to find A^{100} , where $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.
(b) Without expanding prove that $\begin{vmatrix} 0 & -a & -b \\ a & 0 & c \\ b & -c & 0 \end{vmatrix} = 0$.
4. (a) Solve the system of equations by Cramer's rule
$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 14 \\ x - y + z &= 2. \end{aligned}$$

(b) Show that the set of vectors $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is a basis of $\mathbb{R}^3$.

5. (a) Find the cubic roots of 1.
 (b) Solve the equation $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$.
6. (a) Solve the algebraic equation $x^4 + x^2 - 2x + 6 = 0$, it is given that $1 + i$ is a root of the above equation.
 (b) Apply Descartes' rule of sign to examine the nature of the roots of the equation $x^4 + 2x^2 + 3x - 1 = 0$.

Part-B

7. (a) Find the differential equation of all circles which pass through the origin and whose centres are on the y -axis.
 (b) Solve: $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$.
8. (a) Solve: $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$.
 (b) Solve: $(y + \frac{y^3}{3} + \frac{x^2}{2})dx + \frac{(x+xy^2)}{4}dy = 0$.
9. (a) Solve by the method of variation of parameters: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = e^x \tan x$.
 (b) Solve : $(x^2D^2 - xD + 2)y = x \log x$, $(D = \frac{d}{dx})$
10. (a) Find the eigen values and eigen functions of the eigen value problem $\frac{d^2y}{dx^2} + \lambda y = 0$ satisfying with the conditions $y(0) + y'(0) = 0$ and $y(1) + y'(1) = 0$.
 (b) Transform the differential equation $yp^2 + x^3p = x^2y$ ($p = \frac{dy}{dx}$) to Clairaut's form by using the transformation $x^2 = u$ and $y^2 = v$ and hence solve it.
11. (a) Solve by the method of undetermined co-efficients : $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = x^3 - x + e^{-2x}$
 (b) Solve by reducing to normal form $\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + (4x^2 - 3)y = e^{x^2}$.
12. (a) Determine the general solution of the system of differential equations: $\frac{dx}{dt} + 5x + y = e^t$
 $\frac{dy}{dt} + 3y - x = e^{-2t}$.
 (b) Solve by changing the independent variable $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 4\cos(\log(1+x))$.